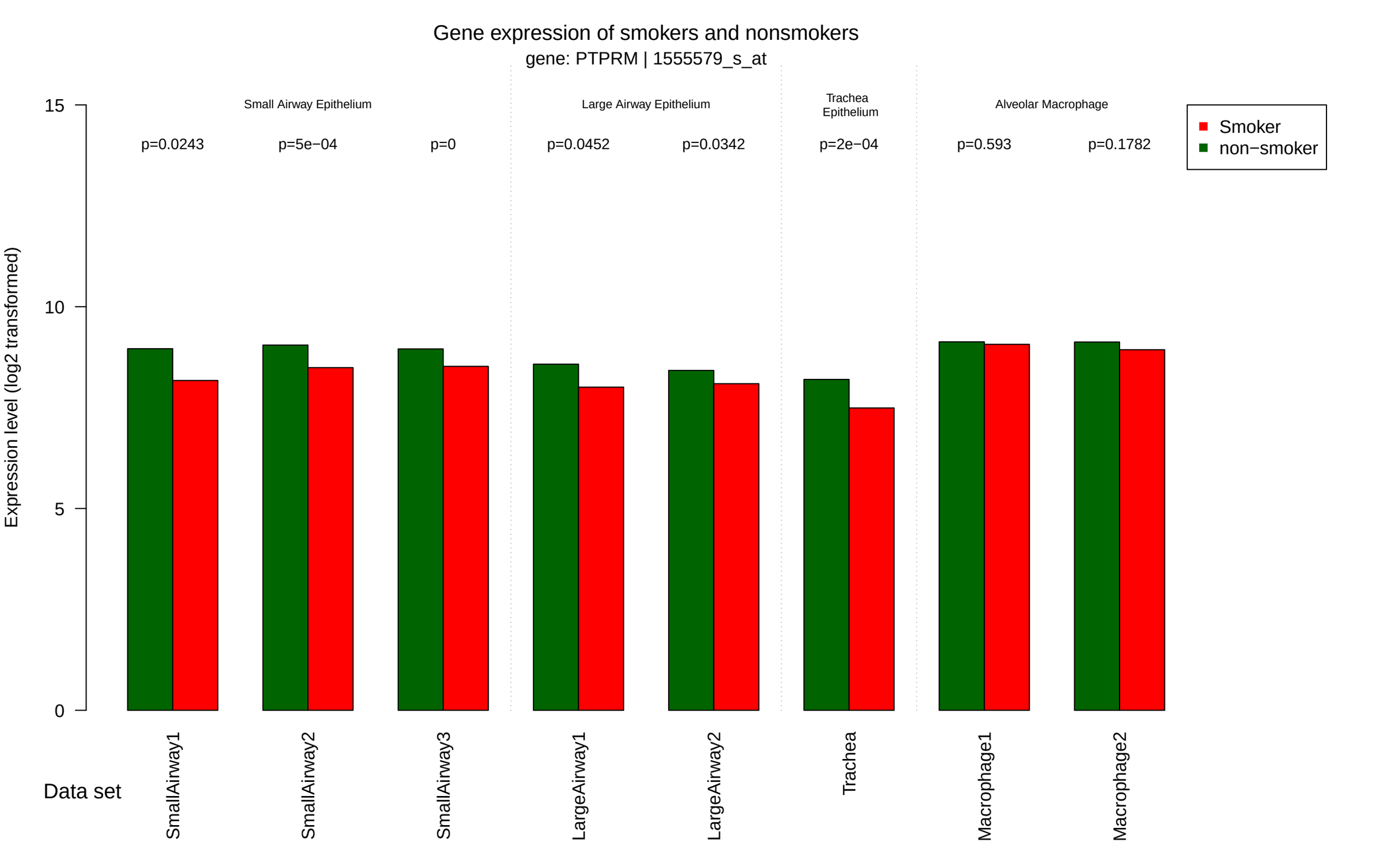
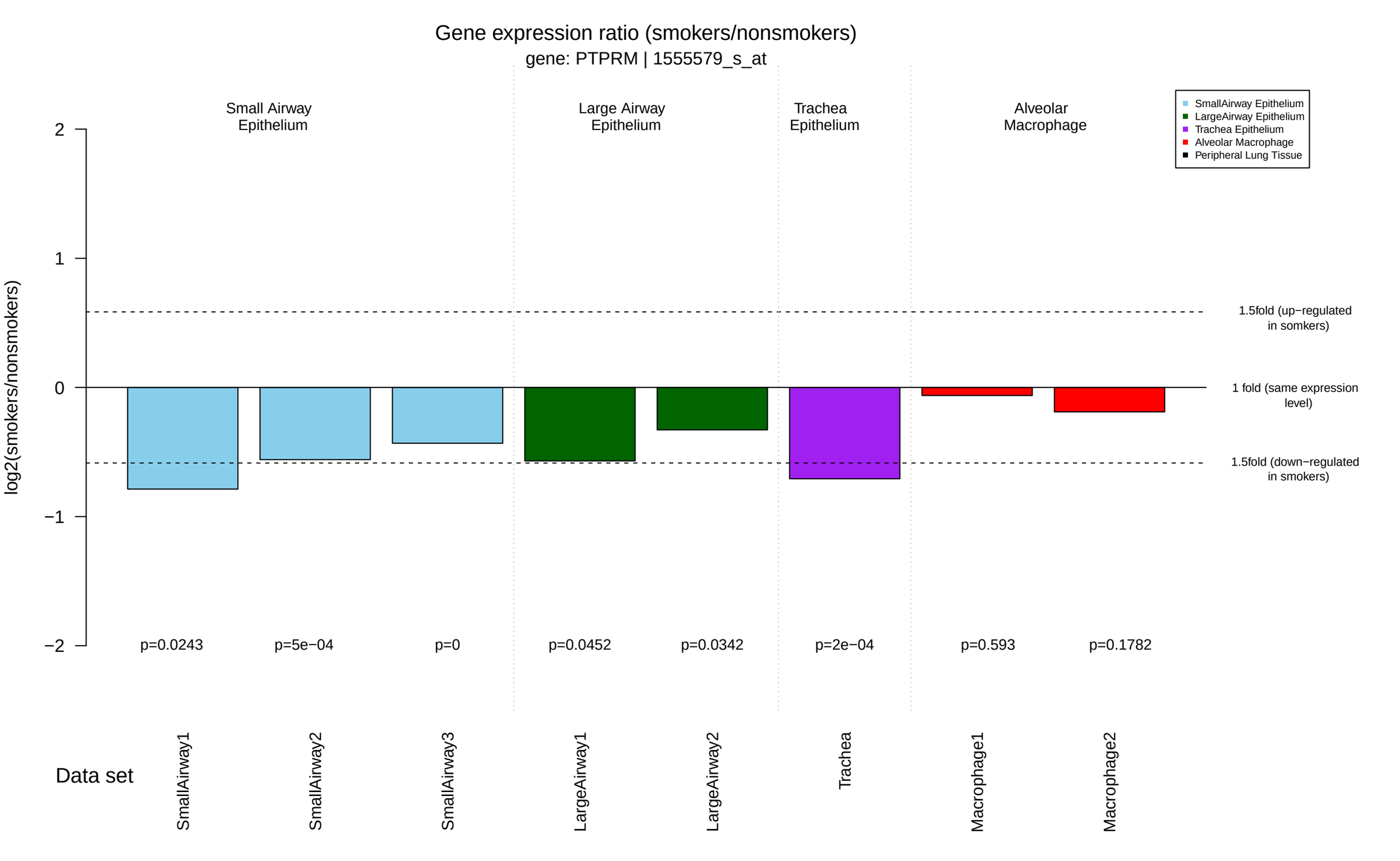
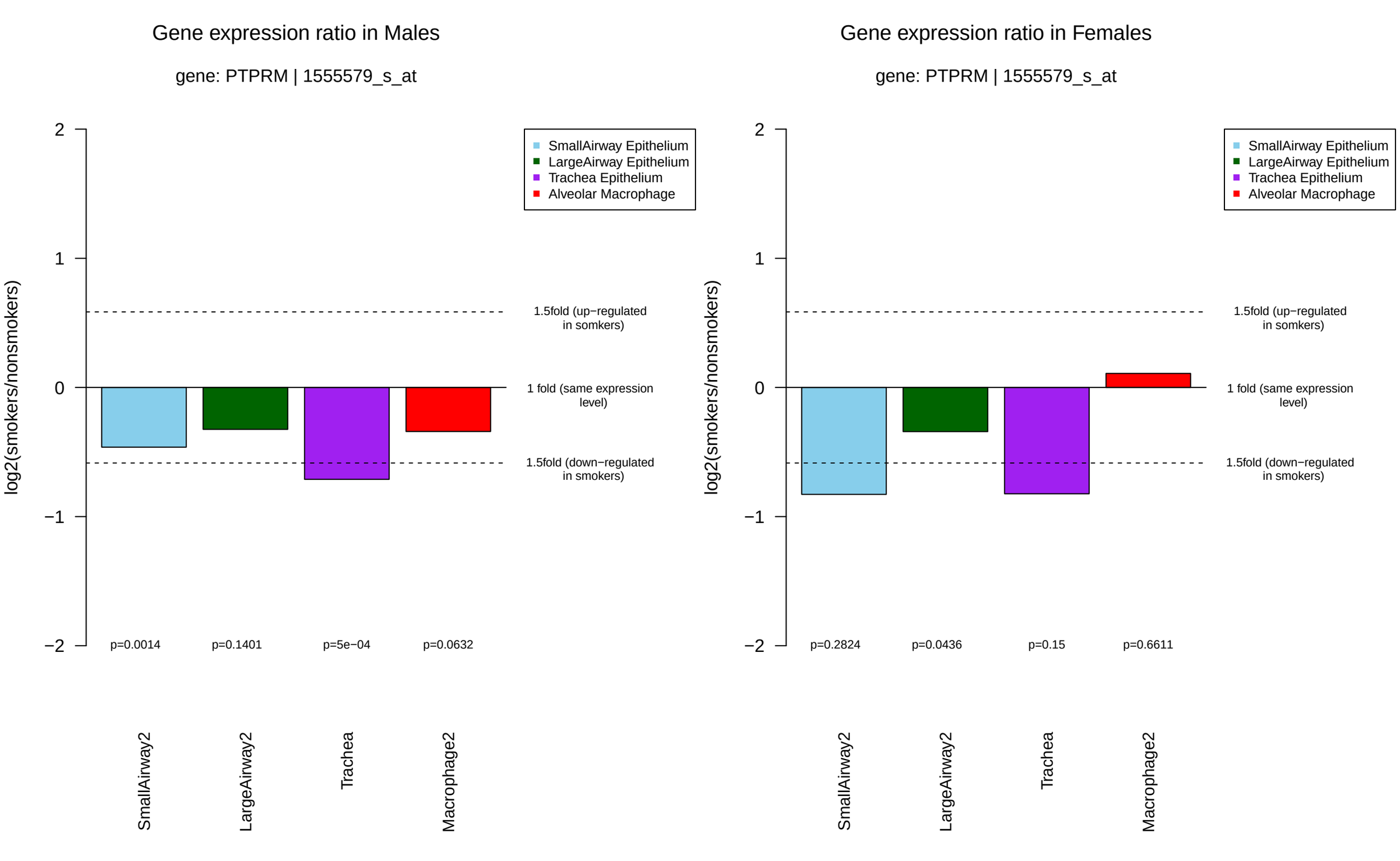
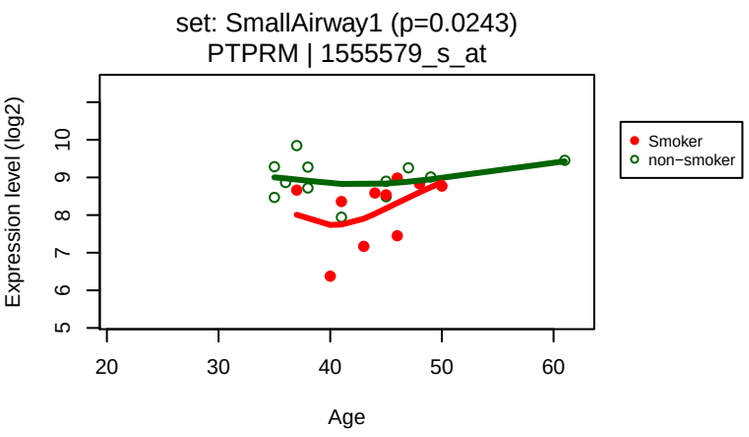
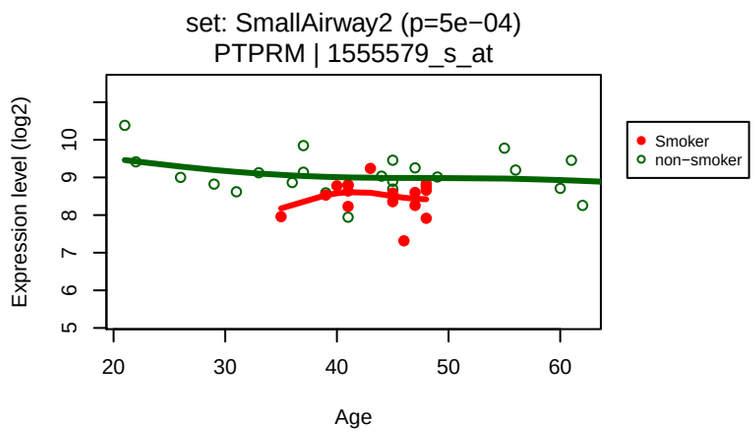
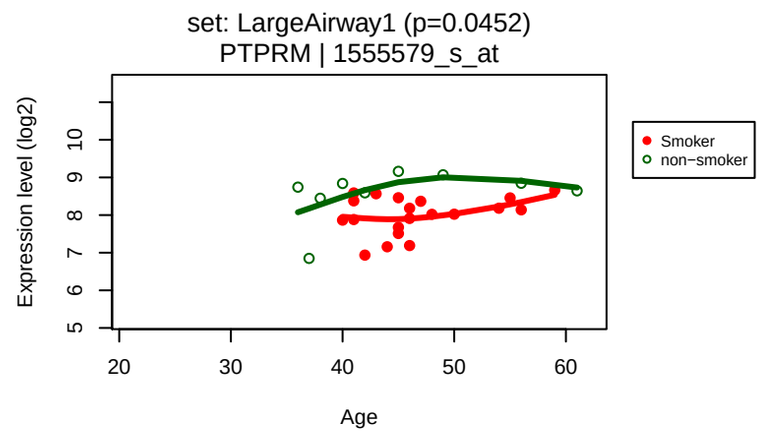

Gene expression of smokers and nonsmokers
gene: PTPRM | 1555579_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0243
p=5e−04
p=0
p=0.0452
p=0.0342
p=2e−04
p=0.593
p=0.1782
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

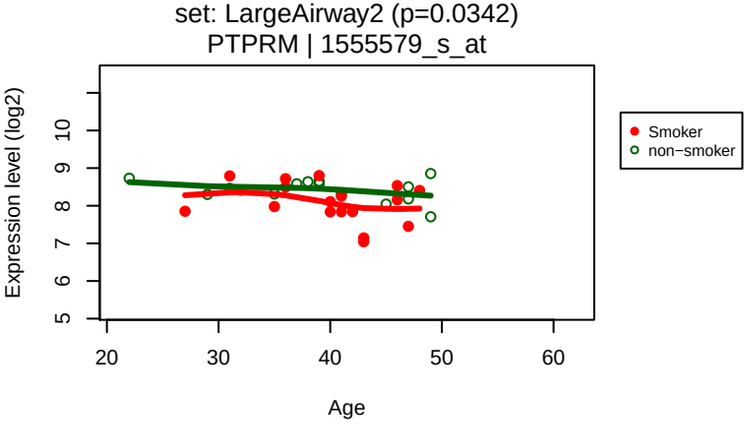

Gene expression ratio (smokers/nonsmokers)
gene: PTPRM | 1555579_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0243
p=5e−04
p=0
p=0.0452
p=0.0342
p=2e−04
p=0.593
p=0.1782
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

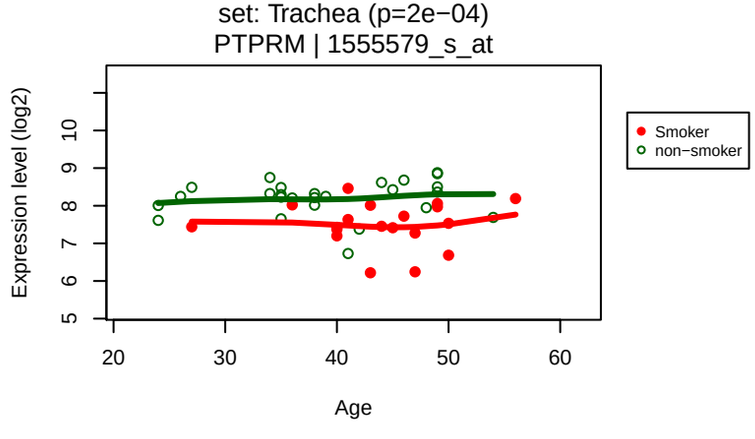

Gene expression ratio in Males
gene: PTPRM | 1555579_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0014
p=0.1401
p=5e−04
p=0.0632
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPRM | 1555579_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2824
p=0.0436
p=0.15
p=0.6611
SmallAirway2
LargeAirway2
Trachea
Macrophage2

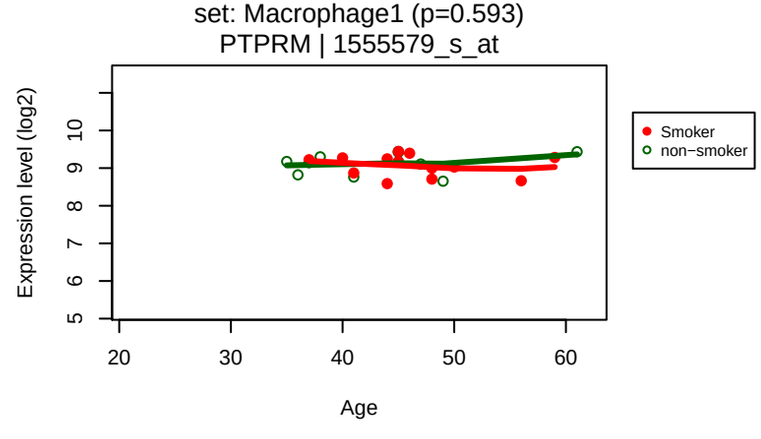

set: SmallAirway1 (p=0.0243)
PTPRM | 1555579_s_at
Expression level (log2)
Age
Smoker
non−smoker

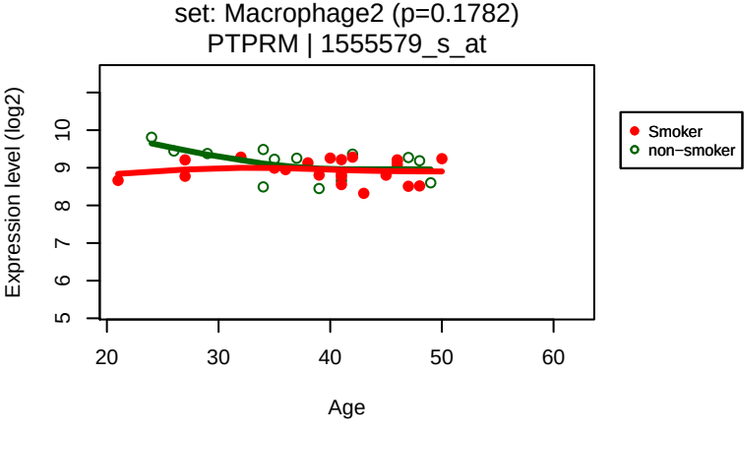

set: SmallAirway2 (p=5e−04)
PTPRM | 1555579_s_at
Expression level (log2)
Age
Smoker
non−smoker

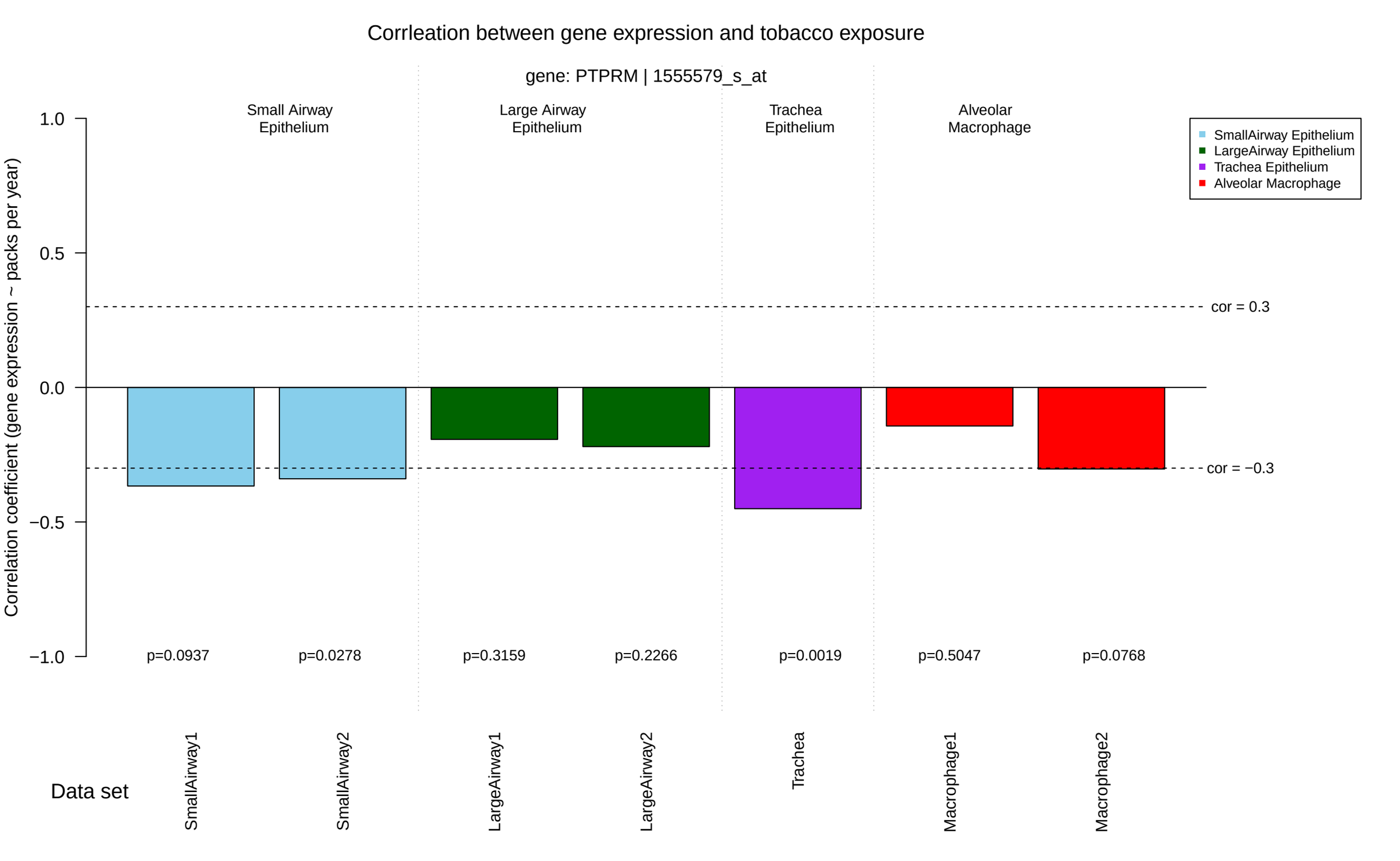

set: LargeAirway1 (p=0.0452)
PTPRM | 1555579_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0342)
PTPRM | 1555579_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
PTPRM | 1555579_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.593)
PTPRM | 1555579_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1782)
PTPRM | 1555579_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPRM | 1555579_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0937
p=0.0278
p=0.3159
p=0.2266
p=0.0019
p=0.5047
p=0.0768
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2